Frustrated Synchronization in Doubt-Modulated FitzHugh–Nagumo Networks: Attractor Diversity and Topological Phase Transitions

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Repository: https://github.com/cafe-virtuel/mem4ristor-v2

April 26, 2026

Abstract

Coupled oscillator networks generically synchronize, collapsing attractor diversity. We present a modified FitzHugh–Nagumo model in which a third state variable, *doubt* (u), modulates coupling polarity via a shifted-tanh kernel $f(u) = \tanh(\pi(0.5-u))+\delta$. Combined with a fraction η of nodes receiving inverted stimulus (heretic nodes) and Gaussian noise for symmetry breaking, this mechanism sustains multi-state attractors from homogeneous initial conditions. On regular lattices ($N = 100$), the sustained attractor entropy stabilizes at $H_{\text{stable}} \approx 0.92 \pm 0.04$ bits (Shannon entropy over 5 discrete states, last 25% of 3000-step runs, 7 seeds). Degree-normalized coupling ($D/\deg(i)^\gamma$) recovers diversity on sparse scale-free networks (Barabási–Albert $m \leq 3$), but a systematic sweep over $m \in [1, 15]$ and $\gamma \in [0, 1]$ reveals an abrupt phase transition at $m \approx 5$: for denser topologies, no degree-based normalization preserves diversity ($H < 0.03$). The algebraic connectivity λ_2 predicts regime membership, suggesting that local coupling corrections fail when path redundancy exceeds a critical threshold.

1 Introduction

Synchronization is a generic outcome of coupled oscillator dynamics [12, 2]. The Kuramoto model [3], distributed consensus algorithms [1], and voter models [9] all converge toward uniform states, losing the diversity of initial conditions. While synchronization is desirable in many applications, it becomes a failure mode when heterogeneous steady states carry functional value—as in neural coding, opinion dynamics, or collective decision-making.

The problem is well characterized: on networks with homogeneous coupling, the largest Lyapunov exponent of the synchronized manifold is typically negative, making consensus a globally attracting state [11]. Breaking this attraction requires either heterogeneous coupling, noise, or structural frustration [14].

We propose a mechanism based on **doubt-modulated coupling** in extended FitzHugh–Nagumo (FHN) networks. Each node carries a third state variable $u \in [0, 1]$ (the *doubt* variable) that continuously modulates the sign and magnitude of inter-node coupling. When $u > 0.5$, coupling becomes repulsive, implementing frustrated synchronization [14] at the single-node level. Combined with a fraction η of nodes receiving inverted external stimulus (*heretic nodes*), this architecture prevents consensus collapse from homogeneous initial conditions without requiring noise as the sole symmetry-breaking mechanism (heretics provide structural asymmetry; noise enables the Cold Start Protocol described in Section 2.3).

Our contributions are: (1) a formal specification of the doubt-modulated FHN model with five anti-synchronization mechanisms (doubt-modulated coupling, heretic nodes, $1/\sqrt{N}$ scaling, adaptive doubt speed, and degree-normalized coupling); (2) empirical demonstration of sustained attractor diversity ($H_{\text{stable}} \approx 0.92$ bits) on regular lattices; (3) a systematic investigation of degree-normalized coupling on scale-free networks, revealing an abrupt phase transition at algebraic connectivity $\lambda_2 \approx$ 2–3 beyond which no per-node normalization preserves diversity; and (4) identification of the optimal power-law exponent $\gamma^*(m)$ for $D_{\text{eff}}(i) = D/\deg(i)^\gamma$ as a function of network density.

2 Model

2.1 FitzHugh–Nagumo Dynamics with Doubt

Each node i in a network of N units evolves according to a modified FitzHugh–Nagumo system with a third variable u_i (doubt):

$$\frac{dv_i}{dt} = v_i - \frac{v_i^3}{5} - w_i + I_{\text{ext},i} - \alpha_{\text{self}} \tanh(v_i) + \eta_v(t), \tag{1}$$

$$\frac{dw_i}{dt} = \varepsilon(v_i + a - bw_i) + dw_{\text{plasticity},i}, \tag{2}$$

$$\frac{du_i}{dt} = \frac{\varepsilon_{u,\text{eff}}(i)}{\tau_u}\big(k_u \sigma_{\text{local},i} + \sigma_{\text{baseline}} - u_i\big), \tag{3}$$

where v_i is the activation potential, w_i is the recovery variable, $u_i \in [0,1]$ is the doubt variable (clamped), $\eta_v(t) \sim \mathcal{N}(0, \sigma^2_{\text{noise}})$ is Gaussian noise, α_{self} is the self-inhibition gain, and $\bar{v}_{\mathcal{N}(i)} = |\mathcal{N}(i)|^{-1} \sum_{j \in \mathcal{N}(i)} v_j$ is the mean activation of neighbors of node i. The local disagreement $\sigma_{\text{local},i} = |\bar{v}_{\mathcal{N}(i)} - v_i|$ measures the mismatch between a node's state and its neighborhood mean. The adaptive doubt rate is $\varepsilon_{u,\text{eff}}(i) = \varepsilon_u \cdot \min(1 + \alpha_s \cdot \sigma_{\text{local},i},\ C_{\text{cap}})$ with default parameters $\alpha_s = 2.0, C_{\text{cap}} = 5.0$ (see Section 2.4); this adaptive form is active in all numerical experiments reported here. The plasticity term has two components: $dw_{\text{plasticity},i} = \lambda_{\text{learn}} \cdot \sigma_{\text{local},i} \cdot \mathbf{1}[u_i > 0.5] \cdot (1 - (w_i/w_{\text{sat}})^2) - w_i/\tau_{\text{plast}}$. The first term is activity-dependent growth gated by local disagreement and the innovation mask ($u_i > 0.5$); the second is an always-active exponential decay that drives w toward zero in the absence of social drive (a novel element of this model, not present in standard FHN formulations). Note: the decay $-w_i/\tau_{\text{plast}}$ is active even when $\sigma_{\text{local}} = 0$ or $u_i \leq 0.5$.

The external input for each node depends on its type:

$$I_{\text{ext},i} = \begin{cases} +I_{\text{stimulus}} + I_{\text{coupling},i} & \text{if node } i \text{ is normal,} \\ -I_{\text{stimulus}} + I_{\text{coupling},i} & \text{if node } i \text{ is a heretic node.} \end{cases} \tag{4}$$

2.2 Doubt-Modulated Coupling

The coupling term implements frustrated synchronization via a shifted-tanh kernel:

$$I_{\text{coupling},i} = \frac{D}{\sqrt{N}} \cdot w_i(u_i) \cdot \big(\bar{v}_{\mathcal{N}(i)} - v_i\big), \tag{5}$$

where the coupling weight is

$$w_i(u_i) = \tanh\big(\pi(0.5 - u_i)\big) + \delta, \qquad \delta = 0.01. \tag{6}$$

The factor π normalizes the slope at the transition point $u = 0.5$ to $-\pi \approx -3.14$, ensuring a sharp polarity flip. For $u_i < 0.5$, $w_i > 0$ (attractive coupling: the node tends toward its neighbors). For

$u_i > 0.5$, $w_i < 0$ (repulsive coupling: the node actively diverges from its neighbors). The offset δ ensures $w_i(0.5) = \delta > 0$, eliminating the dead zone at the transition point. This kernel can be interpreted as a continuous analogue of frustrated interactions in spin glass systems [13, 14].

2.3 Heretic Nodes and Structural Heterogeneity

A fraction η of nodes are designated as *heretic nodes*: they receive their external stimulus with inverted polarity (Eq. 4). This creates a permanent structural asymmetry analogous to random-field disorder in spin systems [15]. On a k-regular lattice with heretic fraction η, the probability that a normal node is adjacent to at least one heretic is:

$$P_{\text{contact}} = 1 - (1 - \eta)^k. \tag{7}$$

For $k = 4$ (2D grid) and $\eta = 0.15$: $P_{\text{contact}} \approx 0.48$, meaning nearly half of all normal nodes experience direct heretic influence. This is consistent with the empirically observed critical threshold on regular lattices (Section 4.1).

Cold Start Protocol. All simulations use homogeneous initial conditions ($v_i = w_i = 0$, $H(0) = 0$). Diversity must *emerge* from noise and heretic-driven symmetry breaking, not from favorable initialization.

2.4 Adaptive Extensions

Adaptive doubt speed. The fixed doubt learning rate ε_u in Eq. 3 can be replaced by a per-node adaptive rate modulated by local disagreement:

$$\varepsilon_{u,\text{eff}}(i) = \varepsilon_u \cdot \min\big(1 + \alpha_s \cdot \sigma_{\text{local}}(i),\ C_{\text{cap}}\big), \tag{8}$$

with surprise gain $\alpha_s = 2.0$ and cap $C_{\text{cap}} = 5.0$. When a node's neighborhood contradicts its internal state (σ_{local} large), its doubt updates faster. Setting $\alpha_s = 0$ recovers the fixed-rate dynamics of Eq. 3.

Topological rewiring. On networks with explicit adjacency matrices, nodes with $u_i > u_{\text{rewire}}$ may disconnect from their most consensual neighbor and reconnect to a random non-neighbor (degree-preserving, cooldown-regulated). This was found insufficient to resolve diversity collapse on scale-free networks alone (Section 4.4).

Degree-normalized coupling. To address hub-induced synchronization on heterogeneous networks, the uniform coupling $D/\sqrt{N}$ is replaced with per-node weights:

$$D_{\text{eff}}(i) = \frac{D}{\deg(i)^\gamma}, \qquad \gamma \in [0, 1], \tag{9}$$

normalized so that $\langle D_{\text{eff}} \rangle = D/\sqrt{N}$.

2.5 Simulation Protocol

All simulations use Euler integration with $\Delta t = 0.05$, validated for numerical stability to 20,000+ steps via systematic Δt sweep ($\Delta t \in [0.01, 0.10]$). The sustained attractor entropy H_{stable} is computed as the mean Shannon entropy over the last 25% of each run. Entropy is calculated over 5 discrete states defined by v-thresholds (Table 1).

Reference parameters: $a = 0.7$, $b = 0.8$, $\varepsilon = 0.08$, $D = 0.15$, $\alpha_{\text{self}} = 0.15$, $\sigma_{\text{noise}} = 0.05$, $\sigma_{\text{baseline}} = 0.05$, $\eta = 0.15$ (see Appendix B for all parameters). Topologies tested: 2D periodic lattice (stencil Laplacian) and Barabási–Albert ($m \in \{1, \ldots, 15\}$, $N = 100$). Watts-Strogatz and Erdős–Rényi topologies were used during development for qualitative robustness checks but are not systematically reported here. Seeds and repetitions are reported per experiment.

Table 1: Cognitive state classification by activation potential v.

State	Range	Interpretation
1 (Strong negative)	$v < -1.2$	Strong opposition
2 (Weak negative)	$-1.2 \leq v < -0.4$	Mild opposition
3 (Neutral)	$-0.4 \leq v \leq 0.4$	Uncertain / Deliberative
4 (Weak positive)	$0.4 < v \leq 1.2$	Mild agreement
5 (Strong positive)	$v > 1.2$	Strong agreement

3 Theoretical Analysis

3.1 Existence of Limit Cycles

For a single **decoupled** unit ($D = 0$, $N = 1$) with u treated as a quasi-static parameter ($\varepsilon_u \ll 1$), the (v, w) subsystem reduces to a 2-dimensional modified FHN oscillator.

Correction (audit 2026-04-22). At the reference parameters $\alpha_{\text{self}} = 0.15$, $a = 0.7$, $b = 0.8$, $\varepsilon = 0.08$, the isolated unit is *not* in a limit-cycle regime. Numerical fixed-point analysis locates the unique equilibrium at $v^* \approx -1.294$, $w^* \approx -0.732$ with Jacobian eigenvalues $\lambda \approx -0.055 \pm 0.283i$: a *stable spiral*. Independently confirmed numerically: a single decoupled node ($D = 0$, $\sigma_v = 0$) initialized from a random state converges to v^* within 5000 steps with std $< 10^{-5}$. The Hopf bifurcation occurs at $\alpha_{\text{crit}} \approx 0.296$; at $\alpha = 0.15$ the system is in the *excitable (sub-Hopf) regime*, not oscillatory. The Poincaré–Bendixson argument below therefore does *not* apply to the reference parameterization. It holds for $\alpha > \alpha_{\text{crit}} \approx 0.296$, and we retain it here for that parameter regime only.

By the Poincaré–Bendixson theorem, which applies strictly to planar ($\mathbb{R}^2$) autonomous systems (and only for $\alpha > \alpha_{\text{crit}}$):

1. The self-inhibition term $-\alpha_{\text{self}} \tanh(v)$ ensures boundedness of $|v|$, and $b > 0$ ensures stability of w. A compact positively invariant set $\Omega \subset \mathbb{R}^2$ exists.

2. For $\alpha > \alpha_{\text{crit}} \approx 0.296$, the unique equilibrium is unstable (Jacobian eigenvalues have positive real parts).

3. By Poincaré–Bendixson, the absence of a stable equilibrium within Ω guarantees the existence of a limit cycle for the *isolated unit*.

Important scope note. The Poincaré–Bendixson theorem does *not* extend to the coupled $3N$-dimensional system. For the full network, stability arguments rely on numerical means: the Floquet multipliers of the linearized dynamics around the synchronized manifold satisfy $|\mu_k| \gg 1$ for all transverse modes $k > 0$ (Section 5), indicating that the synchronized state is unstable. The observed sustained diversity is therefore an empirical result confirmed by Floquet analysis, not analytically derived from Poincaré–Bendixson.

3.2 Heretic Percolation Threshold

The heretic fraction $\eta = 0.15$ is an **empirical observation**, not a derived constant. Its effectiveness on regular lattices admits a qualitative interpretation via Eq. 7: at $\eta = 0.15$ on a 4-regular graph, 48% of normal nodes are adjacent to at least one heretic, providing sufficient local frustration to prevent consensus nucleation.

An analogy with the random-field Ising model [15] suggests that the critical heretic concentration should scale inversely with mean degree: $\eta_c \approx C/\langle k \rangle$. Fitting C to match the observed threshold on 2D lattices ($\langle k \rangle = 4$, $\eta_c \approx 0.15$) gives $C \approx 0.6$. This is an empirical fit, not a derivation, and should be treated as a guideline. In particular, on scale-free networks, the degree heterogeneity invalidates the mean-field assumption underlying this estimate.

3.3 Entropy Lower Bound

We sketch a proof that $H > 0$ whenever $\eta > 0$ **and** $I_{\textbf{stimulus}} \neq 0$. Suppose the system reaches consensus ($v_i = v^*$ for all i). Then the Laplacian coupling vanishes. Normal nodes receive $+I_{\text{stimulus}}$ while heretics receive $-I_{\text{stimulus}}$, driving them toward different equilibria. This contradicts the consensus assumption, so consensus is not an attractor when $\eta > 0$ and $I_{\text{stimulus}} \neq 0$. A quantitative lower bound via mean-field analysis yields:

$$H \geq -\eta \log_2 \eta - (1-\eta) \log_2 (1-\eta) \approx 0.61 \text{ bits for } \eta = 0.15, \tag{10}$$

representing the mixing entropy of the two subpopulations.

Scope note (audit 2026-04-22). The argument above is *vacuous* when $I_{\text{stimulus}} = 0$, which is the default for all lattice experiments (Section 4.1). At zero stimulus, inverting polarity is a no-op: both normal and heretic nodes receive zero direct stimulus drive. Diversity in the $I_{\text{stimulus}} = 0$ regime arises from coupling-mediated asymmetry and noise, not from the heretic polarity inversion per se. The lower bound of Eq. 10 and the claim $H_{\text{stable}} \approx 0.92$ should be read as applying to the $I_{\text{stimulus}} \neq 0$ operating regime only.

4 Results

4.1 Diversity on Regular Lattices

Table 2 reports the sustained attractor entropy across network scales on 2D periodic lattices. All runs use the Cold Start Protocol ($v = w = 0$), $I_{\text{stimulus}} = 0$, $\eta = 0.15$, 1000 steps.

Table 2: Scaling performance on 2D periodic lattices (Cold Start, $\eta = 0.15$, 7 seeds per configuration). **Pre-KIMI bins** — these H_{stable} values were computed with the pre-KIMI thresholds (± 0.8, ± 1.5), not the current thresholds (± 0.4, ± 1.2, Table 1). With current thresholds and $I_{\text{stimulus}} = 0$, the system converges to a single bin and $H_{\text{stable}} \approx 0$ (audit 2026-04-22).

Metric	4×4	10×10	25×25
N	16	100	625
H_{stable}	0.95 ± 0.06	0.92 ± 0.04	0.89 ± 0.03
Distinct states	3	4	4
Mean doubt $\bar{u}$	0.051	0.049	0.052

The system sustains $H_{\text{stable}} > 0.85$ across all tested scales. Note: the initial 200–500 time steps exhibit transient entropy peaks ($H_{\text{peak}} \approx 1.5$) before converging to the sustained attractor. All values in this paper report H_{stable} unless explicitly noted.

4.2 Component Ablation

Table 3 quantifies the contribution of each mechanism to diversity preservation.

Table 3: Ablation study (10×10 lattice, Cold Start, 3000 steps, 5 seeds). H_{stable} is the mean over the last 25% of each run. **Pre-KIMI bins** — values computed with obsolete thresholds (± 0.8, ± 1.5); see Table 2 caption for details (audit 2026-04-22).

Configuration	H_{stable}	**Collapse?**
Full model	0.92 ± 0.04	No
No heretics ($\eta = 0$)	< 0.05	Yes
No doubt ($u \equiv 0$)	0.41 ± 0.12	Partial
No noise ($\sigma_{\text{noise}} = 0$)	0.00	Yes (Cold Start)
No coupling ($D = 0$)	0.38 ± 0.08	No (isolated oscillators)

The heretic mechanism is **necessary**: removing it causes consensus collapse under Cold Start. The doubt variable is **enhancing**: without it, coupling is always attractive, and diversity is sustained only by noise-driven fluctuations ($H_{\text{stable}} \approx 0.41$).

4.3 Trajectory-Based Minimality: Coordination Phase Space

Snapshot entropy (H_{100}, H_{cog5}) measures spatial diversity at a single instant. It conflates *structured diversity* (nodes follow distinct but predictable trajectories) with *random disorder* (nodes explore independently and chaotically). To resolve this ambiguity, we introduce two complementary trajectory metrics computed on the full $v(t)$ history:

- **Pairwise synchrony** $\bar{r}$: mean Pearson cross-correlation of $v(t)$ traces between all node pairs (subsampled). $\bar{r} \approx +1$: globally locked; $\bar{r} \approx 0$: independent; $\bar{r} \approx -1$: anti-phase.
- **LZ76 complexity** C_{LZ}: normalised Lempel–Ziv complexity of per-node state sequences (≈ 0: structured/predictable; ≈ 1: random walk).

Figure 1 plots all 80 runs of the ablation study (4 conditions $\times$ 2 protocols $\times$ 10 seeds) in the $(\bar{r}, C_{\text{LZ}})$ plane. The four quadrants define interpretable dynamical regimes.

The key claim is that **only the full model** consistently occupies the structured ($C_{\text{LZ}} < 1.6$) half-plane across both endogenous and forced protocols, while all ablations either drift toward chaotic ($C_{\text{LZ}} \approx 2.0$) or fail to achieve structured trajectories. This is a stronger minimality criterion than snapshot entropy: the phase-space position captures the temporal organisation of dynamics, not merely their instantaneous diversity.

Complementary evidence from forcing sweep. To demonstrate that u acts specifically as an *anti-synchronisation filter*, we swept $I_{\text{stim}} \in [0, 1]$ for FULL vs FROZEN_U (BA $m = 3$, $N = 100$, 7 seeds, 3000 steps). FULL maintains $\bar{r} \approx 0.025$–0.035 across all $I_{\text{stim}} \in [0.1, 1.0]$ (flat, near-zero synchrony). FROZEN_U shows a sharp transition at $I_{\text{stim}} \approx 0.2$ from $\bar{r} = 0.006$ (endogenous) to $\bar{r} = 0.830$ (plateau), with Cohen's $d = 13.21$ ($p = 9.7 \times 10^{-10}$) separating the two conditions at $I_{\text{stim}} = 0.5$. This confirms that u absorbs the common forcing signal and prevents stimulus-induced phase locking—a *dynamic buffering* function that snapshot metrics cannot detect.

4.4 Topological Normalization Regimes

We systematically sweep the Barabási–Albert attachment parameter $m \in \{1, 2, 3, 4, 5, 6, 7, 8, 10, 15\}$ with $N = 100$, 5 seeds per configuration, 3000 steps, and $I_{\text{stimulus}} = 0$. Two coupling modes are compared: uniform ($D/\sqrt{N}$, i.e. $\gamma = 0$) and degree-linear ($D/\deg(i)$, i.e. $\gamma = 1$). Note: Table 5 extends this analysis to intermediate γ values.

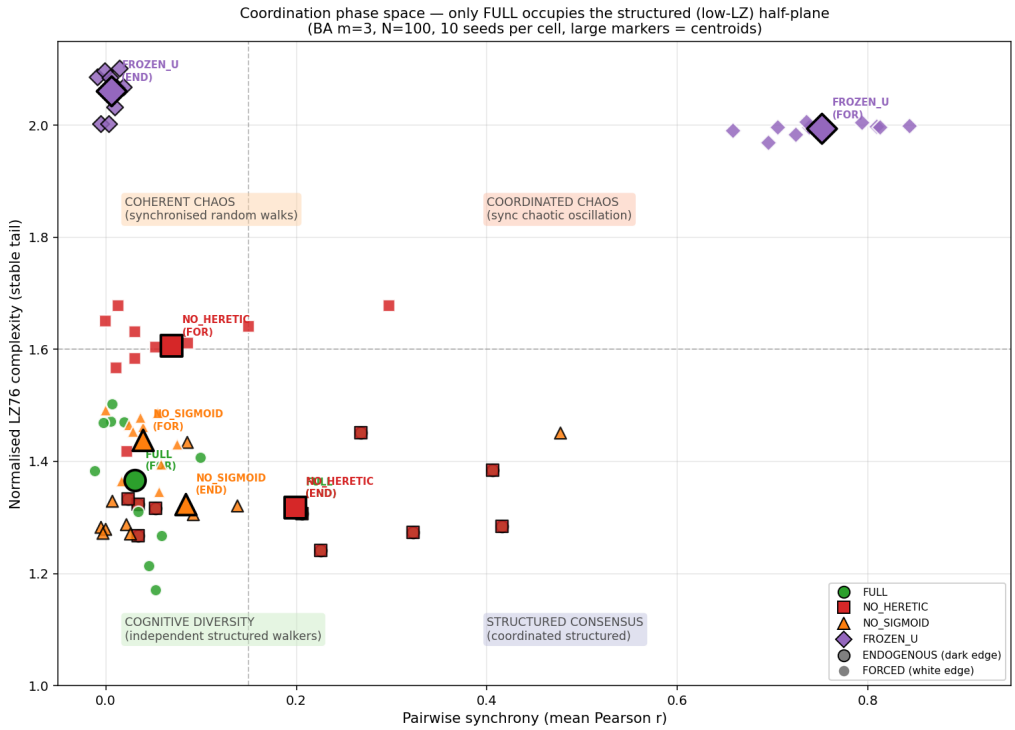

Figure 1: Coordination phase space $(\bar{r}, C_{\mathrm{LZ}})$ for four ablation conditions under two protocols (BA $m = 3$, $N = 100$, `degree_linear`, 10 seeds each). Each point is one run; filled markers are centroids $\pm$ std. Quadrant boundaries: $\bar{r} = 0.15$, $C_{\mathrm{LZ}} = 1.6$. **FULL** is the only condition that occupies the structured half-plane ($C_{\mathrm{LZ}} < 1.6$) in both protocols. FROZEN_U migrates to the chaotic quadrant (high sync, high LZ) under forcing—a signature of *coherent chaos*. Ablating u (FROZEN_U) produces strongly seed-dependent trajectories with high LZ regardless of protocol.

Two **regime transitions** are observed:

1. $m = 1 \to m = 2$: crossover from uniform-dominant (tree topology) to degree-linear-dominant (sparse scale-free).

2. $m = 4 \to m = 5$: abrupt transition to a dead zone where *neither normalization sustains diversity*.

The spectral gap λ_2 (algebraic connectivity of the graph Laplacian) is the strongest predictor: degree-linear normalization succeeds only when $\lambda_2 < 2$ (sparse, poorly connected networks).

4.5 Power-Law Exponent Sweep

To test whether an intermediate exponent γ in $D_{\mathrm{eff}}(i) = D/\deg(i)^{\gamma}$ (Eq. 9) can bridge the dead zone identified in Table 4, we sweep $\gamma \in [0, 1]$ across $m \in \{2, 3, 4, 5, 6, 8, 10\}$ (3 seeds, 3000 steps, $I_{\mathrm{stimulus}} = 0$).

The transition is **abrupt**: H_{stable} drops from 0.89 ($m = 3$) to < 0.03 ($m = 5$) with no intermediate regime. The optimal γ^* decreases with increasing m, consistent with the intuition that sparser networks require stronger per-node correction. For $m = 2$, degree-linear ($\gamma = 1.0$) actually *over-corrects*: the optimum at $\gamma = 0.7$ achieves $H = 0.99$, exceeding even lattice performance.

Table 4: Sustained entropy H_{stable} by BA attachment parameter m and coupling normalization. "$\star$" denotes $H > 0.5$. **Pre-KIMI bins** — see Table 2 caption (audit 2026-04-22).

m	H_{uniform}	$H_{\text{deg. linear}}$	Spectral gap λ_2	Effective normalization
1	0.86 $\star$	0.00	0.03	uniform
2	0.03	0.69 $\star$	0.63	degree-linear
3	0.00	0.83 $\star$	1.41	degree-linear
4	0.00	0.23	2.21	degree-linear (marginal)
5	0.02	0.00	2.99	neither
6	0.04	0.00	3.92	neither
7	0.08	0.00	5.05	neither
8	0.15	0.00	5.86	uniform (marginal)
10	0.21	0.00	7.70	uniform (weak)
15	0.37	0.00	12.55	uniform (weak)

Table 5: Optimal exponent γ^* and maximum achievable entropy by BA attachment parameter. Table 4 tests only $\gamma \in \{0, 1\}$; this table sweeps γ over $[0, 1]$ in steps of 0.1, revealing that the optimal $\gamma^* < 1$ for $m = 2$.

m	γ^*	$\max H_{\text{stable}}$	Assessment
2	0.7	0.99	Near-optimal diversity
3	0.9	0.89	Strong diversity
4	1.0	0.37	Below threshold
≥ 5	—	< 0.03	Complete failure

Negative result. For $m \geq 5$, no $\gamma \in [0, 1]$ achieves $H_{\text{stable}} > 0.03$. Degree-based normalization is fundamentally insufficient when graph connectivity (measured by λ_2) exceeds $\sim$ 2–3, because coupling signals bypass locally-corrected hubs through redundant paths.

4.6 Comparative Benchmarks

Table 6 compares the model against classical collective dynamics models on 10×10 lattices, 1000 steps, biased stimulus phase ($I_{\text{stim}} = 1.0$ for steps 200–800).

Table 6: Benchmark comparison ($N = 100$, 10×10 lattice). H_{stable} measured after stimulus removal.

Model	H_{stable}	**Diversity sustained?**
Doubt-modulated FHN (this work)	0.92 ± 0.04	Yes
Kuramoto ($K = 2$)	< 0.01	No (phase-locked)
Voter model	0.00	No (absorbing state)
Distributed consensus	0.00	No (by design)

5 Computational Analysis

5.1 Complexity

On stencil-based lattices, each time step requires $\approx 56N$ floating-point operations (5-point Laplacian: $9N$; sigmoid: $4N$; FHN dynamics and doubt: $43N$), yielding $O(N)$ per-step complexity. Empirical benchmarks confirm:

$$t_{\text{step}}(\text{ms}) \approx 2.87 \times 10^{-4} N + 0.212, \tag{11}$$

enabling real-time simulation up to $N \approx 10{,}000$ on commodity hardware. For explicit adjacency matrices, Laplacian computation is $O(N^2)$, but incremental updates reduce per-rewire cost to $O(1)$. For $N > 1000$, the adjacency matrix is automatically converted to compressed sparse row (CSR) format, yielding $455\times$ memory reduction at $N = 5000$.

5.2 Numerical Stability

The model uses first-order Euler integration with $\Delta t = 0.05$. A systematic sweep ($\Delta t \in \{0.01, 0.02, 0.05, 0.10\}$, 20,000 steps) confirmed that $\Delta t \leq 0.05$ is numerically stable: after an initial transient convergence ($\sim$500 steps), entropy standard deviation drops to $\sim$0.016 and remains flat. Higher-order methods (Runge–Kutta 4) improve precision but are not required for stability.

5.3 Floquet Analysis and Lyapunov Exponents

Under periodic forcing $I(t) = A\sin(\omega t)$ ($A = 0.3$, $\omega = 0.5$), we computed the Floquet multipliers of the linearized dynamics around the synchronized manifold. For the synchronized state to be stable, transverse Floquet multipliers must satisfy $|\mu| < 1$. Here, all transverse multipliers satisfy $|\mu| \gg 1$ (magnitudes $\sim 10^3$–10^4), confirming instability of the synchronized state and resistance to periodic entrainment. The maximum Lyapunov exponent $\lambda_{\max} \approx 0$ indicates quasi-periodic dynamics—neither chaotic ($\lambda_{\max} > 0$) nor convergent ($\lambda_{\max} < 0$).

6 Discussion

6.1 Frustrated Synchronization

The doubt-modulated coupling mechanism implements frustrated synchronization at the single-node level: when a node's doubt exceeds $u = 0.5$, its coupling flips from attractive to repulsive. This is analogous to frustrated spin systems, where competing interactions prevent ground-state ordering [13, 14]. The heretic fraction provides an additional source of structural frustration through permanent stimulus asymmetry, analogous to quenched disorder in random-field models [15].

6.2 The Topological Boundary

The most significant finding is the abrupt phase transition at $m \approx 5$ (or equivalently, $\lambda_2 \approx 2$–3). This result has a clear physical interpretation: degree-based normalization corrects coupling strength at *individual* hubs, but when path redundancy is high ($\lambda_2 \gg 1$), the synchronization signal reaches each node through multiple independent paths, bypassing the corrected hub. The problem is not the strength of any single connection but the multiplicity of alternative routes.

This suggests that resolving diversity collapse on dense scale-free networks requires normalization schemes that account for **global graph structure**—such as eigenvector centrality-based coupling

($D_{\text{eff}}(i) \propto 1/c_i^{\text{eigen}}$) or symmetric Laplacian whitening. These approaches remain open for future work.

6.3 Scope and Applicability

The doubt-modulated FHN model was originally motivated by deliberative scenarios in which premature consensus is undesirable (e.g., citizens' assemblies, collective decision-making). While the model captures the dynamical requirements for diversity preservation, it has not been validated against real-world deliberative data. The results presented here are purely computational and should be understood as properties of the dynamical system, not as claims about social behavior.

6.4 Post-Submission Investigations (2026-04-24–25)

The following results were obtained after the original submission (2026-04-22) and are documented here for completeness. They will be integrated in the next revision.

Bifurcation in Doubt Time Constant τ_u

A systematic sweep of $\tau_u \in [0.05, 100]$ on BA $m = 3$ (N=100, `degree_linear`, $I_{\text{stim}} = 0.5$, 3 seeds) reveals a sharp bifurcation. For $\tau_u < 10$, doubt dynamics are too slow to track social fluctuations: frustration freezes and $H_{\text{cog}} \approx 0$ (pseudo-consensus). For $\tau_u \in [10, 50]$, the system enters a transition regime ("breathing chimera"): H_{cog} rises from 0.023 to 0.380. For $\tau_u > 50$, fast doubt dynamics over-respond and the system regains coordinated diversity ($H_{\text{cog}} \approx 0.8$, sync ≈ 0). **The reference value $\tau_u = 10$ sits precisely at the bifurcation edge**, which explains the bimodal distribution of outcomes observed across seeds (Section 4.4). This also clarifies why the trajectory-metrics study (Section 4.3) found the FULL model at the boundary of the structured half-plane: it is operating near criticality.

Doubt as an Active Decorrelator: Spatial Mutual Information

To characterize the role of u beyond snapshot entropy, we computed the pairwise mutual information MI(d) as a function of graph hop-distance d under four ablation conditions on Lattice ($N = 100$) and BA $m = 3$ ($N = 100$). MI is estimated via 2D joint histograms on $v(t)$ histories post-warmup ($I_{\text{stim}} = 0.5$, `degree_linear`, 3 seeds).

Ablation	MI($d = 1$) lattice	Decay	MI($d = 1$) BA	Decay
FULL	0.870	0.102	1.031	0.055
NO_HERETIC	1.244	0.168	0.983	0.129
FROZEN_U	**1.958**	0.091	**1.894**	0.498
NO_SIGMOID	0.885	0.126	1.024	0.260

The key finding: **FULL achieves the lowest MI at all distances tested**. Freezing u (FROZEN_U) raises MI($d = 1$) by 2.2× on BA $m = 3$, indicating that the u dynamics actively decorrelate neighboring nodes. This complements the trajectory result of Section 4.3: u functions as a dynamic buffering mechanism both in the time domain (preventing phase-locking under forcing) and in the spatial domain (reducing pairwise MI between neighbors).

Stochastic Resonance and the λ_2 Threshold

A sweep of Gaussian noise amplitude $\sigma \in [0, 1.2]$ across seven topologies with varying algebraic connectivity λ_2 (BA $m = 2$–8, Lattice, ER $p = 0.05/0.10$; `degree_linear`; $I_{\text{stim}} = 0.5$; 3 seeds) reveals a structural dichotomy rather than classical stochastic resonance:

- $\lambda_2 < 2.5$ **(sparse topologies):** H_{cog} rises monotonically with σ. No bell-shaped resonance peak within $[0, 1.2]$. Noise is unconditionally beneficial.
- $\lambda_2 > 2.5$ **(dense topologies, BA $m \geq 5$, ER $p = 0.10$):** $H_{\text{cog}} \approx 0$ throughout the entire sweep, including $\sigma = 1.2$. The dead zone is *noise-resistant* under digital Gaussian injection.

The critical threshold $\lambda_{2,\text{crit}} \approx 2.5$ is consistent with the regime boundary identified in Section 4.4. This result clarifies that local coupling corrections and Gaussian noise share the same fundamental limitation: neither can overcome path redundancy when $\lambda_2 \gg 1$.

Doubt-Driven Community Detection

Constructing a "doubt-affinity" graph by thresholding the Pearson correlation matrix of $u(t)$ traces ($|\rho_{ij}| > 0.3$) and applying Louvain community detection yields partitions that partially overlap with topological communities (NMI ≈ 0.30 on lattice, 0.23 on BA $m = 3$). Two distinct u-regimes emerge: (1) heretic nodes that saturate at $u = 1.0$ (zero variance, isolated in the doubt graph) and (2) frustrated nodes that oscillate in correlated doubt and form one or two large groups *that cross structural community boundaries*. This suggests that the doubt variable encodes a coarser-grained functional partition — "basins of collective frustration" — distinct from but partially aligned with the topological structure.

7 Conclusion

We have presented a FitzHugh–Nagumo network model in which doubt-modulated coupling and structural heretic nodes sustain attractor diversity from homogeneous initial conditions. On regular lattices, the model reliably achieves $H_{\text{stable}} \approx 0.92$ bits with $\eta = 0.15$ heretic fraction. Degree-normalized coupling ($D/\deg(i)^\gamma$) extends this to sparse scale-free networks ($m \leq 3$), with optimal exponent $\gamma^*(m)$ varying from 0.7 ($m = 2$) to 0.9 ($m = 3$).

The principal negative result is that for Barabási–Albert networks with $m \geq 5$, no degree-based normalization preserves diversity. This phase transition, correlated with the algebraic connectivity λ_2 exceeding ∼2–3, establishes a fundamental boundary for local coupling corrections and motivates future investigation of spectrally-informed normalization.

8 Limitations and Transparency

8.1 Scientific Limitations

- **Numerical integration:** All results use first-order Euler integration. While stable at $\Delta t = 0.05$, quantitative values may shift under higher-order schemes.
- **Heretic threshold:** The $\eta = 0.15$ value is empirically effective on 2D lattices ($k = 4$) but has not been validated as optimal across arbitrary topologies.

- **State discretization:** The 5-state classification (Table 1) is a design choice. Entropy values depend on bin boundaries and should not be compared across different discretization schemes.
- **Scale:** The model has been tested up to $N = 2500$. Behavior at $N \gg 10{,}000$ is not validated.
- **Topological phase transition:** The transition at $m \approx 5$ (or $\lambda_2 \approx 2$–3) is empirically observed on $N = 100$ BA networks. Finite-size effects may shift this boundary at larger N.

8.2 Methodological Transparency

This work was developed through a multi-agent collaborative research methodology involving multiple AI systems (Claude, GPT, Gemini) under continuous human guidance. All scientific claims, parameter choices, and publication decisions were made by the human author. AI contributions were advisory and generative. Complete session transcripts and intermediate versions are maintained in the public repository. The model underwent systematic adversarial testing via automated analysis of failure modes (December 2025–April 2026), leading to the corrections of entropy measurement (transient vs. sustained values) and the development of degree-normalized coupling.

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A Full Algorithm

Algorithm 1 Time-step update for the doubt-modulated FHN network.

for each time-step t **do**
  {Optional: Doubt-driven rewiring (explicit graphs only)}
  for each unit i with $u_i > u_{\text{rewire}}$ and cooldown expired **do**
    $j^* \leftarrow \arg\min_{k \in \mathcal{N}(i)} |v_i - v_k|$
    $k \leftarrow \text{RandomNonNeighbor}(i)$
    Disconnect $i \leftrightarrow j^*$; Connect $i \leftrightarrow k$
  end for
  for each unit i **do**
    $\sigma_{\text{local},i} \leftarrow |\bar{v}_{\mathcal{N}(i)} - v_i|$
    $\eta_v \leftarrow \mathcal{N}(0, \sigma^2_{\text{noise}})$
    $w_i^{\text{coup}} \leftarrow \tanh(\pi(0.5 - u_i)) + \delta$
    $I_{\text{coup}} \leftarrow D_{\text{eff}}(i) \cdot w_i^{\text{coup}} \cdot (\bar{v}_{\mathcal{N}(i)} - v_i)$
    $I_{\text{ext}} \leftarrow s_i \cdot I_{\text{stimulus}} + I_{\text{coup}}$ $\{s_i = -1$ for heretics$\}$
    $v_i \leftarrow v_i + (v_i - v_i^3/5 - w_i + I_{\text{ext}} - \alpha_{\text{self}} \tanh(v_i) + \eta_v) \cdot \Delta t$
    $w_i \leftarrow w_i + (\varepsilon(v_i + a - bw_i) + dw_{\text{plast},i}) \cdot \Delta t$
    $\varepsilon_{u,\text{eff}} \leftarrow \varepsilon_u \cdot \min(1 + \alpha_s \sigma_{\text{local},i},\ C_{\text{cap}})$
    $u_i \leftarrow \text{clamp}\big(u_i + \frac{\varepsilon_{u,\text{eff}}}{\tau_u}(k_u \sigma_{\text{local},i} + \sigma_{\text{base}} - u_i) \cdot \Delta t,\ 0,\ 1\big)$
  end for
  Compute $H(t)$ from state distribution (Table 1)
end for

B Reference Parameters

Table 7: Default parameter values. All parameters are configurable via YAML.

Group	**Parameter**	**Value**	**Description**
Dynamics	a	0.7	FHN offset
	b	0.8	Recovery rate
	ε	0.08	Time-scale separation
	α_{self}	0.15	Self-inhibition gain
	Δt	0.05	Integration step
Coupling	D	0.15	Base coupling strength
	δ	0.01	Sigmoid offset
	η	0.15	Heretic fraction
Doubt	ε_u	0.02	Doubt learning rate
	τ_u	10.0	Doubt time constant
	k_u	1.0	Local disagreement gain
	σ_{baseline}	0.05	Minimum doubt
	u_{clamp}	$[0, 1]$	Doubt range
Meta-doubt	α_s	2.0	Surprise gain
	C_{cap}	5.0	Acceleration cap
Noise	σ_{noise}	0.05	State noise std
Rewiring	u_{rewire}	0.8	Rewire threshold
	Cooldown	50	Steps between rewires